

Lab 1 – RedFlag Product Description

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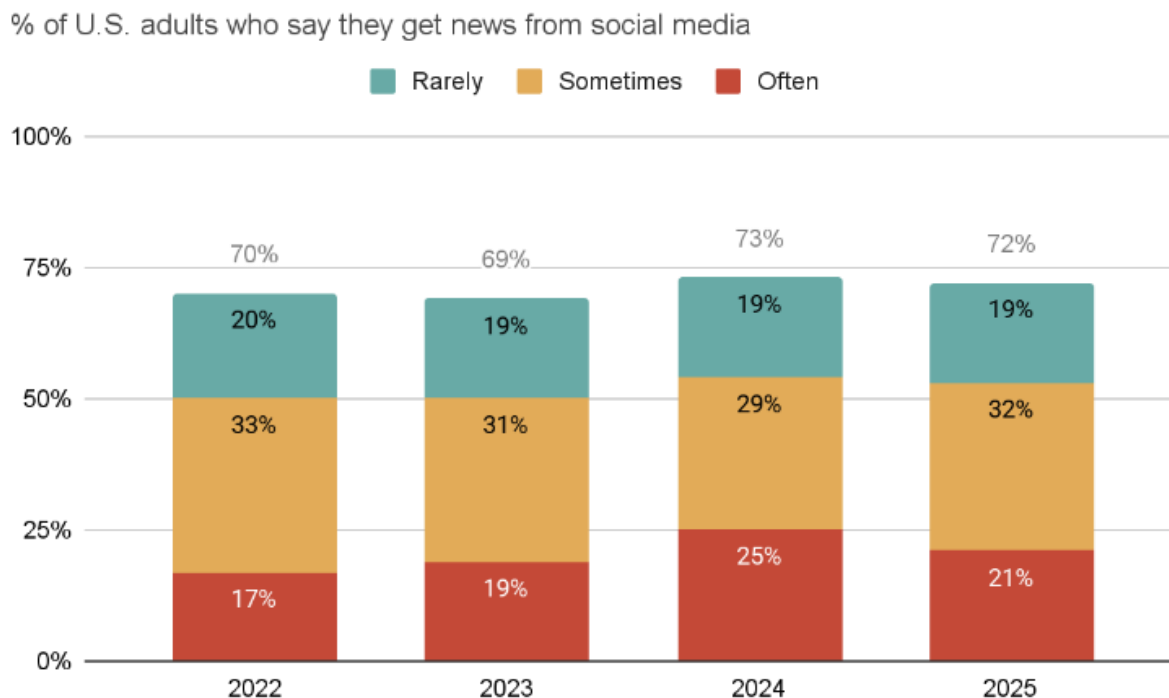
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1 Introduction

Misinformation now spreads through social media platforms faster and further than accurate reporting and information does. A large-scale analysis of 11 years of Twitter data found that while true stories rarely reach 1,000 people, false stories reach between 1,000 and 100,000 people, while also diffusing faster than the truth (Vosoughi et al., 2018). This matters more and more each year because social media has become the most dominant source of news; just over half of U.S. adults now regularly get their news from social media since 2022 (Pew Research Center, 2025).

Figure 1

% of US adults who say they get news from social media



Much of this spread of misinformation is not intentional. Researchers studying why this happens have found that people share misinformation less because they cannot recognize it and

more because they are simply not paying attention to accuracy while scrolling (Ordway, 2021). Social media platforms have also been compounding the problem. Due to the fact that likes, shares, and comments are all methods for users to interact with posts as well as reward mechanisms for the creators of the content (regardless of the truthfulness of the content), users develop a sharing behavior that pays little attention to accuracy (Ceylan, 2023).

As social media becomes the primary source of news for more individuals, the need for a tool to discern between factual information and misinformation at the moment it is viewed or shared becomes more important. The tool to close this gap is RedFlag.

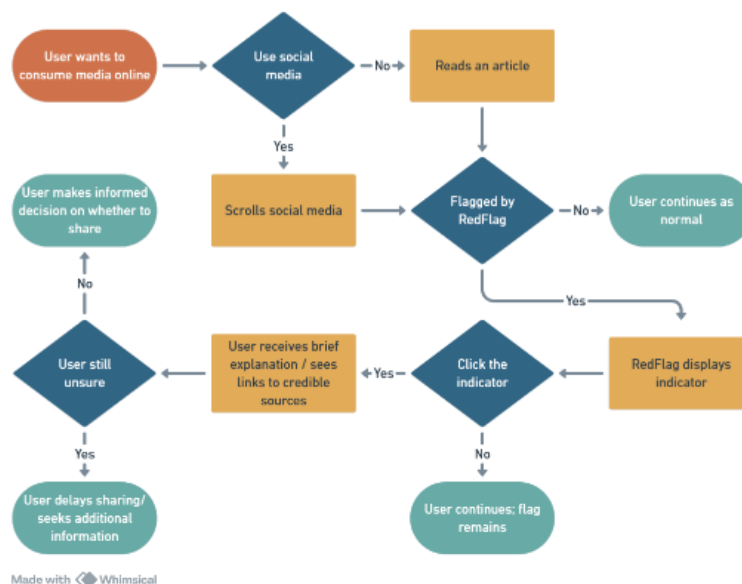
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2 Product Description

RedFlag is a browser extension designed to close the gap for a product that discerns between factual information and misinformation at the time of viewing. Rather than requiring users to leave their browsing experience or manually seek out a fact-checking tool, RedFlag runs passively in the background as users scroll through social media and other websites, scanning on-screen text and highlighting content that may be false or misleading. Through clicking on this highlighted text, users can view credible sources so that they can evaluate the claim for themselves, if they choose to do so. Another feature, BiasStar, flags content on politically divisive topics rather than blatant misinformation and provides sources representing multiple viewpoints across the political spectrum, helping users recognize one-sided content rather than telling them what to believe.

Figure 2

RedFlag High Level Concept Diagram



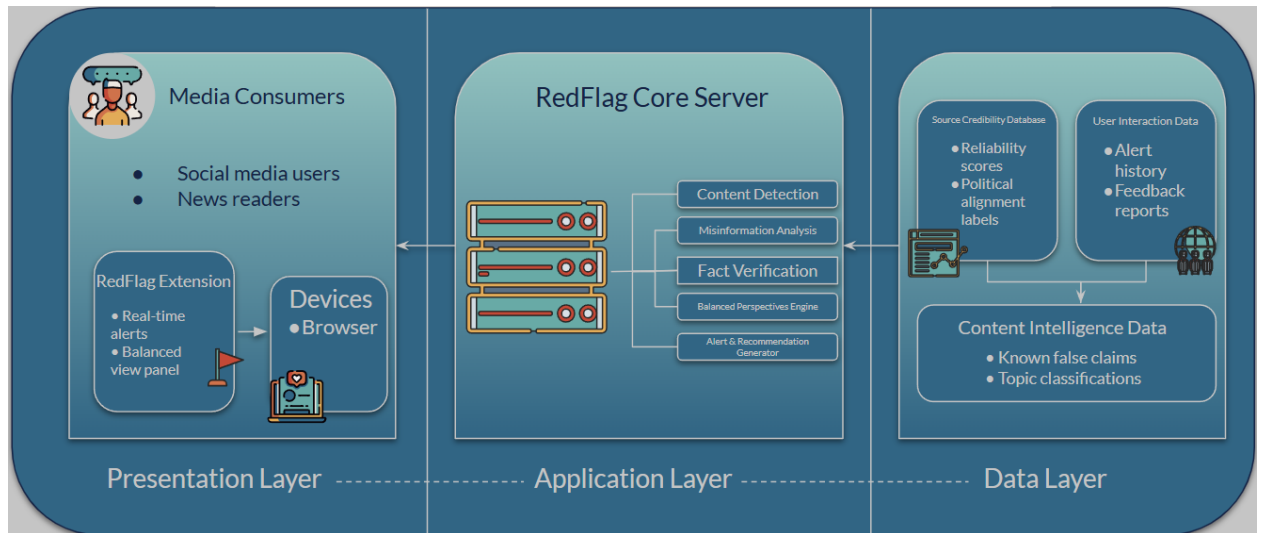
2.1 Key Product Features and Capabilities

RedFlag's core capability is real-time integrated detection, which scans a user's screen as they scroll through websites to identify potentially false or misleading claims. This detection runs automatically in the background and requires no action from the user, such that RedFlag never interrupts the normal flow of browsing to demand input (such as alternatively forcing the user to close a pop-up). By working with the user's news consumption flow instead of against it, RedFlag avoids the inconvenience of checking sources elsewhere – one of the largest issues faced in the fight against misinformation today.

For content on politically divisive topics, RedFlag's Balanced Perspectives Engine, aptly named BiasStar, marks the content with a star and offers sources representing separate viewpoints across the political spectrum. Instead of attaching a correct or incorrect label to content, BiasStar offers various perspectives so that users can decide for themselves which viewpoint is correct. This feature is intended to counter the tendency of social media algorithms to reinforce a more extreme single point of view for its users by making alternative credible viewpoints easy to find.

2.2 Major Components (Hardware/Software)

RedFlag is designed to run on any modern desktop or laptop computer capable of running a Chromium or Firefox-based browser. The system is organized into three layers consistent with the Major Functional Component Diagram (MFCD) shown in Figure 3, those being a presentation layer, application layer, and data layer, supported by a set of external APIs.

Figure 3*Major Functional Component Diagram (MFCD)*

The presentation layer is the RedFlag browser extension itself, the only component users directly interact with, also known as the front end. It is responsible for displaying highlight alerts on flagged content and for rendering the expandable view panel that shows sources and explanations when a user requests them. The front end of the software is built primarily with JavaScript and communicates with the backend to retrieve detection results and source information as the user browses.

The application layer, or back end, is the processing hub of RedFlag and hosts the system's core logic. It contains five primary components: content detection (which extracts and prepares on-screen text for analysis), misinformation analysis (which evaluates that text against known claims and credibility signals), fact verification (which checks flagged claims against external sources), the Balanced Perspectives Engine (which enables BiasStar's multi-viewpoint recommendations), and the recommendation generator (which packages the results into highlights and explanations finally shown to the user). This layer will be built with Python, chosen for its multitude of libraries for handling and processing large data sets.

The data layer will be built in PostgreSQL and will store three categories of information: a source credibility database containing reliability scores and political-alignment labels for known sources; user interaction data; and previously identified false claims and topic classifications. This central set of data allows RedFlag to evaluate new content quickly by comparing it to prior findings rather than starting from scratch with each scan.

3 Identification of Case Study

RedFlag is designed for everyday social media users who want to avoid misinformation without slowing down to fact-check every post they deem suspicious.

The representative target user is Daisuke, a 16-year-old who spends much of his free time scrolling on Instagram. He often notices that many posts he views seem inaccurate, but manually researching each one takes more time and effort than he is willing to spend, so most of what he sees goes unchecked.

This user's experience reflects the core problem RedFlag aims to solve: users are aware misinformation exists in their feed and generally know how to check this information online, but the cost of leaving the platform to verify a claim discourages them from doing so with any amount of consistency. With RedFlag enabled in his browser, Daisuke's experience changes. Questionable text is automatically highlighted as he scrolls, and clicking the indicator brings him directly to credible sources, letting him check a claim in seconds instead of minutes. Because RedFlag requires no extra effort beyond installing it once, Daisuke is likely to keep it running and recommend it to friends who scroll the same platforms, making word of mouth a natural path to adoption.

Although Daisuke represents RedFlag’s primary case study, the same need extends beyond a single demographic; just one social media platform, Facebook, has users in their teens and their 70s, both male and female from all across the world. Any casual or engaged social media user faces the same tradeoff between speed and accuracy, so RedFlag’s low-effort design is meant to appeal to this wide audience.

Table 1

Table of Comparison between RWP and Prototype

Feature	Prototype	Real World Product
Fact-checking data source	Live connections to external factchecking and news APIs	Same as Prototype
Browser support	Chrome or Firefox only	Chrome, Firefox, and Safari
Detection scope	Static text on a large set of supported sites	All major platforms and text analysis within images
BiasStar source coverage	Small hand-picked list of sources	Continuously updated source-credibility database
Explanations	LLM generated explanations for any flagged claim, with sources given from our database	Quicker responses than Prototype

4 Glossary

API: Application Programming Interface

BiasStar: An optional feature of RedFlag that highlights politically divisive information rather than just misinformation

Credibility Score: A percentage score given to a source that evaluates its credibility / trustworthiness

Misinformation: False or misleading information

NLP: Natural Language Processing; the understanding, interpretation, and generation of natural human language by computers

RAG: Retrieval-Augmented Generation, a technique used to improve the accuracy and reliability of LLMs by fetching facts from external sources before generating a response.

5 References

Vosoughi, S., Roy, D., & Aral, S. (2018). *The spread of true and false news online*.

<https://doi.org/10.1126/science.aap9559>

Pew Research Center. (2025, September 25). *Social media and news fact sheet*.

<https://www.pewresearch.org/journalism/fact-sheet/social-media-and-news-fact-sheet/>

Ordway, D. (2021, March 23). *Why do Americans share so much fake news? One big reason is they aren't paying attention, new research suggests*. Nieman Journalism Lab.

<https://www.niemanlab.org/2021/03/why-do-americans-share-so-much-fake-news-one-big-reason-is-they-arent-paying-attention-new-research-suggests/>

Ceylan, G. (2023, March 31). *How social media rewards misinformation*. Yale Insights.

<https://insights.som.yale.edu/insights/how-social-media-rewards-misinformation>